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TD N° 4: Administration Systèmes Informatiques

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Objectif

L'objectif de ce TP est de **maîtriser l'identification des ports ouverts, la vérification des services actifs et la configuration des pare-feu (UFW, iptables et nftables)**, ainsi que le contrôle du service SSH et de Fail2Ban, afin d'assurer la **sécurité et la protection d'un serveur Linux**.

1. Analyse du serveur

- Identifier les ports ouverts:** `ss -tuln` ou `netstat -tuln`
- Vérifier les services actifs :** `systemctl list-units --type=service`

```
evenort12@ubuntu:~$ ss -tuln
Netid State Recv-Q Send-Q Local Address:Port  Peer Address:Port Process
udp    UNCONN 0      0      127.0.0.54:53      0.0.0.0:*
udp    UNCONN 0      0      127.0.0.53:10:53   0.0.0.0:*
udp    UNCONN 0      0      192.168.56.101:3:68 0.0.0.0:*
udp    UNCONN 0      0      0.0.0.0:111        0.0.0.0:*
udp    UNCONN 0      0      127.0.0.1:161      0.0.0.0:*
udp    UNCONN 0      0      [::]:111           [::]:*
udp    UNCONN 0      0      [::1]:161          [::]:*
tcp    LISTEN 0      4096    0.0.0.0:10050      0.0.0.0:*
tcp    LISTEN 0      4096    127.0.0.53:10:53   0.0.0.0:*
tcp    LISTEN 0      4096    127.0.0.54:53      0.0.0.0:*
tcp    LISTEN 0      100     0.0.0.0:25         0.0.0.0:*
tcp    LISTEN 0      4096    0.0.0.0:22         0.0.0.0:*
tcp    LISTEN 0      4096    0.0.0.0:111        0.0.0.0:*
tcp    LISTEN 0      511     0.0.0.0:80         0.0.0.0:*
tcp    LISTEN 0      4096    [::]:10050         [::]:*
tcp    LISTEN 0      100     [::]:25            [::]:*
tcp    LISTEN 0      4096    [::]:22            [::]:*
tcp    LISTEN 0      4096    [::]:111           [::]:*

evenort12@ubuntu:~$ systemctl list-units --type=service
UNIT
● apache2.service
● apparmor.service
● appport.service
● blk-availability.service
● console-setup.service
● cron.service
● dbus.service
● finalrd.service
● getty@tty1.service
● keyboard-setup.service
● kmod-static-nodes.service
● lvm2-monitor.service
● ModemManager.service
● multipathd.service
● nagios4.service
● networkd-dispatcher.service
● nginx.service
● plymouth-quit-wait.service
● plymouth-quit.service
● plymouth-read-write.service
● polkit.service
```

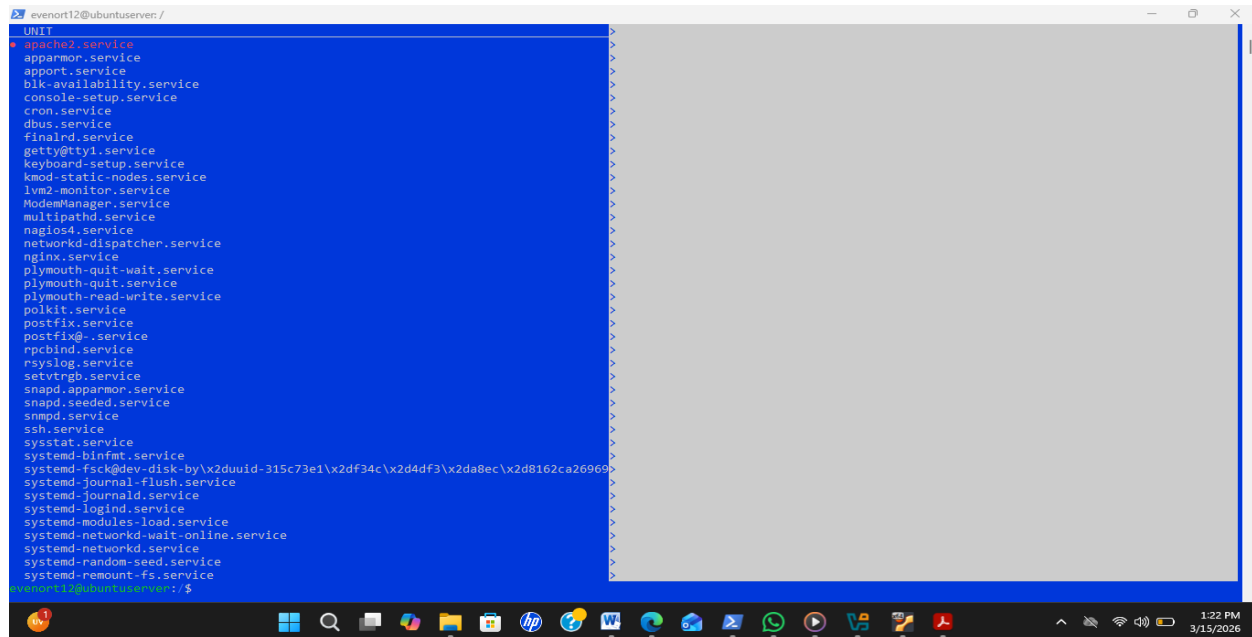


Figure 1, 2: Ces images présentent l'identification des ports ouverts ainsi que la vérification des services actifs, confirmant que l'analyse du serveur a été réalisée avec succès (voir pages 2 à 3).

2. Firewall simple avec UFW

- a) Installer UFW : `sudo apt install ufw -y`
- b) Vérifier l'état du firewall : `sudo ufw status`
- c) Politique par défaut

`sudo ufw default deny incoming`

`sudo ufw default allow outgoing`

```
evenort12@ubuntu:~$ sudo apt install ufw -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ufw est déjà la version la plus récente (0.36.2-6).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 43 non mis à jour.
evenort12@ubuntu:~$ sudo ufw status
Status: inactive
evenort12@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
evenort12@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
evenort12@ubuntu:~$
```

Figure 3: Cette image confirme que les paquets UFW ont été installés et que l'état du service firewall a été vérifié avec succès.

2.1 Autoriser les services nécessaires

```
sudo ufw allow 22
sudo ufw allow 80
sudo ufw allow 443
```

2.2 Activer le firewall

```
sudo ufw enable
```

a) Vérifier :

```
sudo ufw status verbose
```

```
evenont12@ubuntu:~$ sudo apt install ufw -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ufw est déjà la version la plus récente (0.36.2-6).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 43 non mis à jour.
evenont12@ubuntu:~$ sudo ufw status
Status: inactive
evenont12@ubuntu:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
evenont12@ubuntu:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
evenont12@ubuntu:~$ sudo ufw allow 22
Rules updated
Rules updated (v6)
evenont12@ubuntu:~$ sudo ufw allow 80
Rules updated
Rules updated (v6)
evenont12@ubuntu:~$ sudo ufw allow 443
Rules updated
Rules updated (v6)
evenont12@ubuntu:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
evenont12@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
443 ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)

evenont12@ubuntu:~$
```

Figure 4: Cette image illustre l'autorisation des services nécessaires et confirme que la vérification en mode verbose a été réalisée avec succès.

a) Afficher les règles actuelles : `sudo iptables -L`

b) Bloquer tout le trafic entrant : `sudo iptables -P INPUT DROP`

c) Autoriser SSH : `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`

d) Autoriser HTTPS : `sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT`

e) Autoriser les connexions déjà établies

`sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED`

```
evenort12@ubuntu:~$ sudo iptables -L
Chain INPUT (policy DROP)
target prot opt source destination
ufw-before-logging-input all -- anywhere anywhere
ufw-before-input all -- anywhere anywhere
ufw-after-input all -- anywhere anywhere
ufw-after-logging-input all -- anywhere anywhere
ufw-reject-input all -- anywhere anywhere
ufw-track-input all -- anywhere anywhere

Chain FORWARD (policy DROP)
target prot opt source destination
ufw-before-logging-forward all -- anywhere anywhere
ufw-before-forward all -- anywhere anywhere
ufw-after-forward all -- anywhere anywhere
ufw-after-logging-forward all -- anywhere anywhere
ufw-reject-forward all -- anywhere anywhere
ufw-track-forward all -- anywhere anywhere

Chain OUTPUT (policy ACCEPT)
target prot opt source destination
ufw-before-logging-output all -- anywhere anywhere
ufw-before-output all -- anywhere anywhere
ufw-after-output all -- anywhere anywhere
ufw-after-logging-output all -- anywhere anywhere
ufw-reject-output all -- anywhere anywhere
ufw-track-output all -- anywhere anywhere

Chain ufw-after-forward (1 references)
target prot opt source destination

Chain ufw-after-input (1 references)
target prot opt source destination
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:netbios-ns
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:netbios-dgm
ufw-skip-to-policy-input tcp -- anywhere anywhere tcp dpt:netbios-ssn
ufw-skip-to-policy-input tcp -- anywhere anywhere tcp dpt:microsoft-ds
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:bootps
ufw-skip-to-policy-input udp -- anywhere anywhere udp dpt:bootpc
ufw-skip-to-policy-input all -- anywhere anywhere ADDRTYPE match dst-type BROADCAST

Chain ufw-after-logging-forward (1 references)
target prot opt source destination
```

```
evenort12@ubuntu:~$ sudo iptables -L
target prot opt source destination limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "
LOG all -- anywhere anywhere

Chain ufw-after-logging-input (1 references)
target prot opt source destination limit: avg 3/min burst 10 LOG level warn prefix "[UFW BLOCK] "
LOG all -- anywhere anywhere

Chain ufw-after-logging-output (1 references)
target prot opt source destination

Chain ufw-after-output (1 references)
target prot opt source destination

Chain ufw-before-forward (1 references)
target prot opt source destination
ACCEPT all -- anywhere anywhere ctstate RELATED,ESTABLISHED
ACCEPT icmp -- anywhere anywhere icmp destination-unreachable
ACCEPT icmp -- anywhere anywhere icmp time-exceeded
ACCEPT icmp -- anywhere anywhere icmp parameter-problem
ACCEPT icmp -- anywhere anywhere icmp echo-request
ufw-user-forward all -- anywhere anywhere

Chain ufw-before-input (1 references)
target prot opt source destination
ACCEPT all -- anywhere anywhere ctstate RELATED,ESTABLISHED
ACCEPT all -- anywhere anywhere ctstate INVALID
ufw-logging-deny all -- anywhere anywhere
DROP all -- anywhere anywhere
ACCEPT icmp -- anywhere anywhere icmp destination-unreachable
ACCEPT icmp -- anywhere anywhere icmp time-exceeded
ACCEPT icmp -- anywhere anywhere icmp parameter-problem
ACCEPT icmp -- anywhere anywhere icmp echo-request
ACCEPT udp -- anywhere anywhere udp spt:bootps dpt:bootpc
ufw-not-local all -- anywhere anywhere
ACCEPT udp -- anywhere anywhere 224.0.0.251 udp dpt:mdns
ACCEPT udp -- anywhere anywhere 239.255.255.250 udp dpt:1900
ufw-user-input all -- anywhere anywhere

Chain ufw-before-logging-forward (1 references)
target prot opt source destination

Chain ufw-before-logging-input (1 references)
target prot opt source destination
```

```
evenort12@ubuntu:~/server/
Chain ufw-track-input (1 references)
target    prot opt source                destination
Chain ufw-track-output (1 references)
target    prot opt source                destination
ACCEPT    tcp  --  anywhere              anywhere        ctstate NEW
ACCEPT    udp  --  anywhere              anywhere        ctstate NEW
Chain ufw-user-forward (1 references)
target    prot opt source                destination
Chain ufw-user-input (1 references)
target    prot opt source                destination
ACCEPT    tcp  --  anywhere              anywhere        tcp dpt:ssh
ACCEPT    udp  --  anywhere              anywhere        udp dpt:22
ACCEPT    tcp  --  anywhere              anywhere        tcp dpt:http
ACCEPT    udp  --  anywhere              anywhere        udp dpt:80
ACCEPT    tcp  --  anywhere              anywhere        tcp dpt:https
ACCEPT    udp  --  anywhere              anywhere        udp dpt:https
Chain ufw-user-limit (0 references)
target    prot opt source                destination
LOG        all  --  anywhere              anywhere        limit: avg 3/min burst 5 LOG level warn prefix "[UFW LIMIT BLOCK] "
REJECT     all  --  anywhere              anywhere        reject-with icmp-port-unreachable
Chain ufw-user-limit-accept (0 references)
target    prot opt source                destination
ACCEPT    all  --  anywhere              anywhere
Chain ufw-user-logging-forward (0 references)
target    prot opt source                destination
Chain ufw-user-logging-input (0 references)
target    prot opt source                destination
Chain ufw-user-logging-output (0 references)
target    prot opt source                destination
Chain ufw-user-output (1 references)
target    prot opt source                destination
evenort12@ubuntu:~/server/
```

Figure 5, 6, 7: Ces images illustrent l’affichage des règles actuelles à l’aide de la commande **sudo iptables -L**.

```
evenort12@ubuntu:~/server/
evenort12@ubuntu:~/server/$ sudo iptables -P INPUT DROP
evenort12@ubuntu:~/server/$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
evenort12@ubuntu:~/server/$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
evenort12@ubuntu:~/server/$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
evenort12@ubuntu:~/server/$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
evenort12@ubuntu:~/server/$
```

Figure 8: Cette image présente les autorisations accordées pour différents ports, configurées à l’aide de la commande **iptables**.

a) **Installation** : `sudo apt install nftables -y`

- b) **Activer** : `sudo systemctl enable nftables`
- c) **Éditer (configuration)** : `sudo nano /etc/nftables.conf`
- d) **sudo systemctl restart nftables**

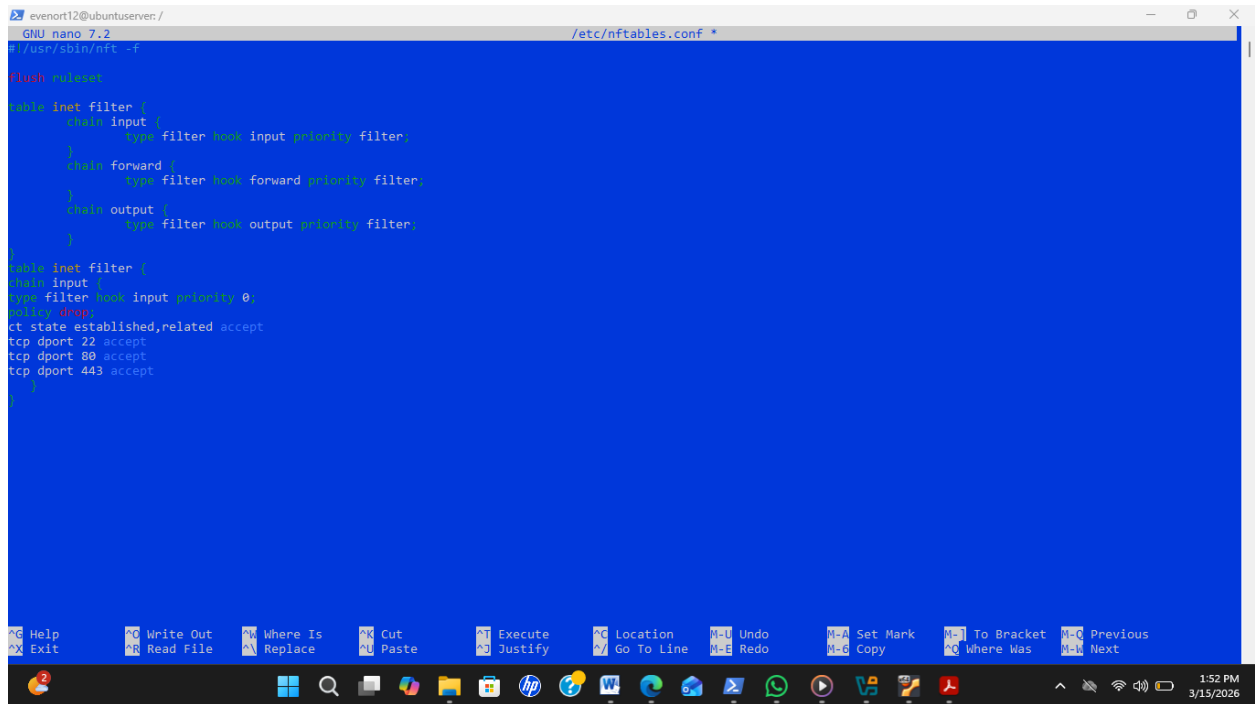


Figure 9 : Cette image montre l'édition du fichier de configuration à l'aide de la commande `sudo nano /etc/nftables.conf`.

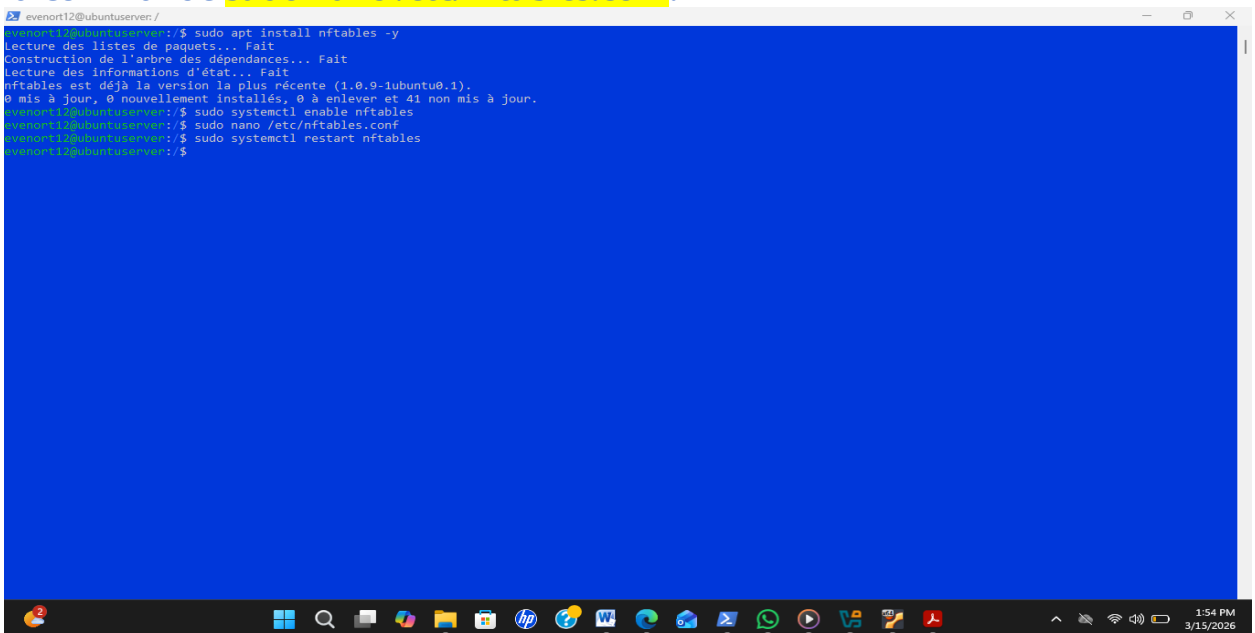
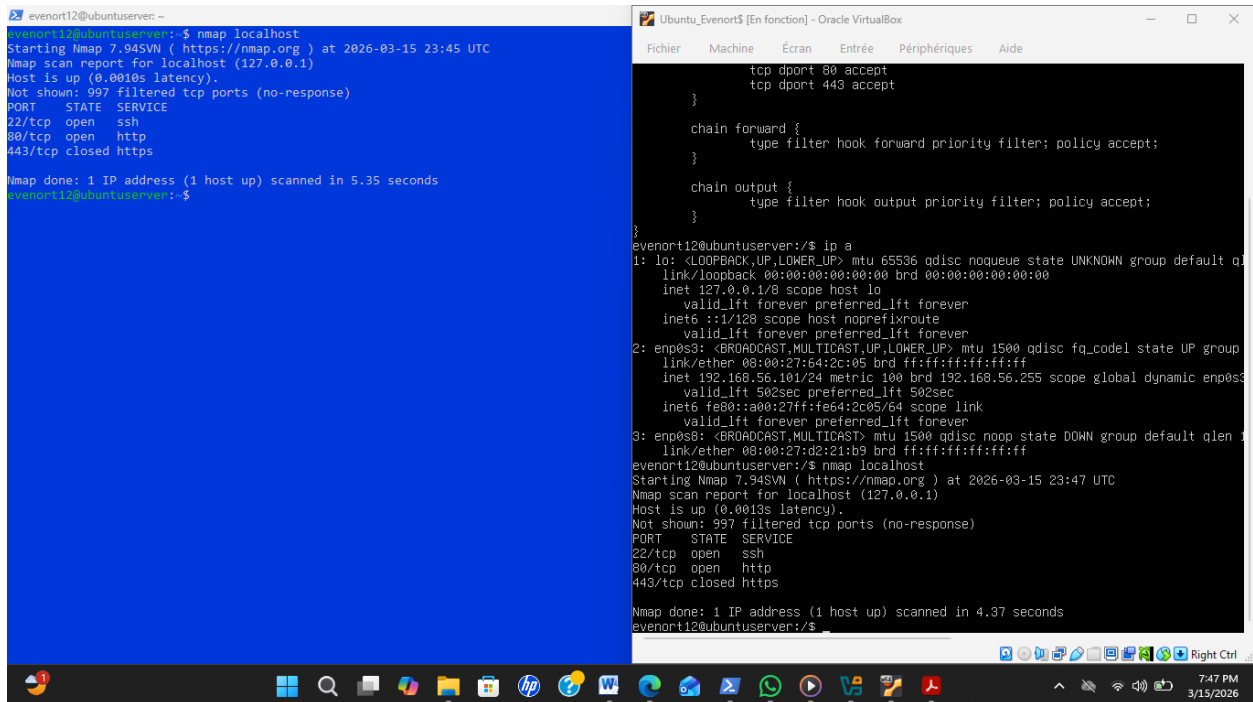


Figure 10: Cette image confirme l'installation des `paquets nftables` ainsi que le redémarrage du `service nftables`, assurant son bon fonctionnement.

4. Tests de sécurité

a) Tester les ports ouverts : nmap localhost



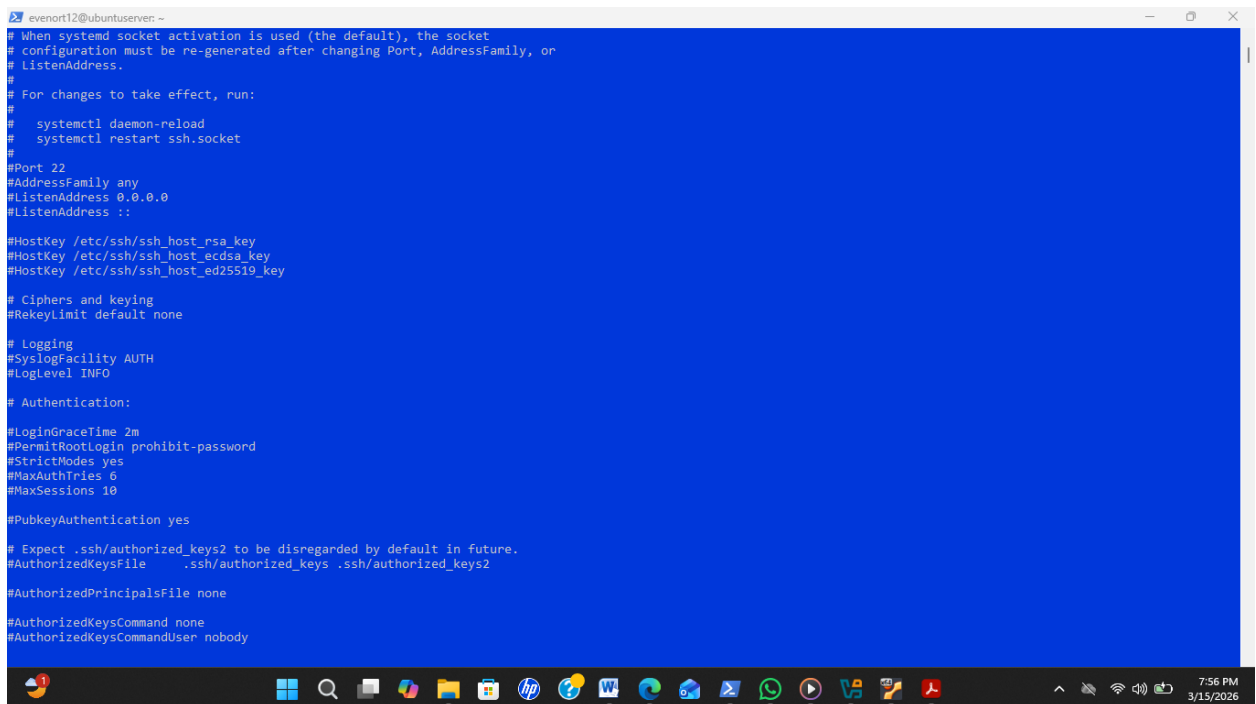
```
evenort12@ubuntu:~$ nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-15 23:45 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0010s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 5.35 seconds
evenort12@ubuntu:~$
```

```
evenort12@ubuntu:~$ nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-15 23:47 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0013s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 4.37 seconds
evenort12@ubuntu:~$
```

Figure 11: Cette image montre que les tests de sécurité ont été réalisés avec succès.



```
evenort12@ubuntu:~$ cat /etc/ssh/sshd_config
# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
#
#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
#
# Ciphers and keying
#RekeyLimit default none
#
# Logging
#SyslogFacility AUTH
#LogLevel INFO
#
# Authentication:
#
#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
#PubkeyAuthentication yes
#
# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2
#
#AuthorizedPrincipalsFile none
#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
```

```
evenort12@ubuntu:~$ cat /etc/ssh/sshd_config

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes
```

```
evenort12@ubuntu:~$ cat /etc/ssh/sshd_config

# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
evenort12@ubuntu:~$
```

Figure 12, 13, 14 : Ces images confirment que les ports ouverts ont été analysés avec succès à l'aide de la commande **nmap localhost** (voir pages 8 à 9).

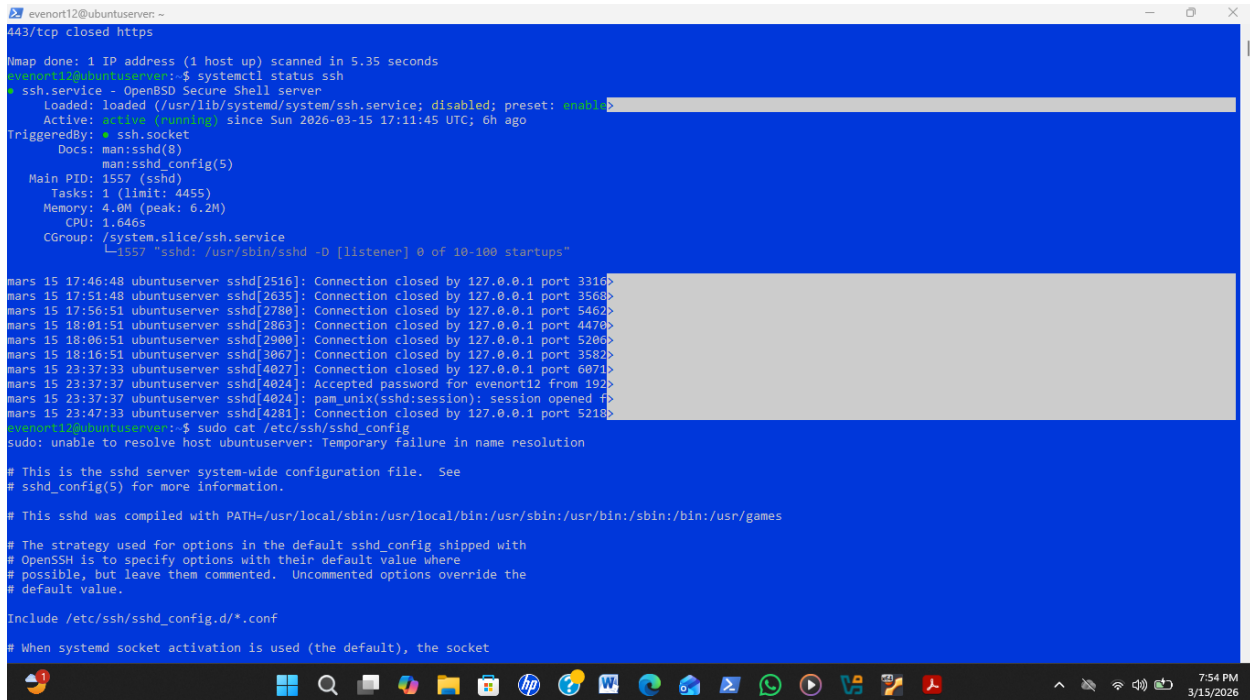
Durcissement d'un serveur OpenSSH

6. Analyse de la configuration actuelle

a) Afficher la configuration SSH :

systemctl status ssh

sudo cat /etc/ssh/sshd_config



```
evenort12@ubuntu:~$ systemctl status ssh
443/tcp closed https
Nmap done: 1 IP address (1 host up) scanned in 5.35 seconds
evenort12@ubuntu:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Sun 2026-03-15 17:11:45 UTC; 6h ago
   TriggeredBy: * ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
   Main PID: 1557 (sshd)
   Tasks: 1 (limit: 4455)
   Memory: 4.0M (peak: 6.2M)
   CPU: 1.646s
   CGroup: /system.slice/ssh.service
           └─1557 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

mars 15 17:46:48 ubuntu: sshd[2516]: Connection closed by 127.0.0.1 port 3310
mars 15 17:51:48 ubuntu: sshd[2635]: Connection closed by 127.0.0.1 port 3568
mars 15 17:56:51 ubuntu: sshd[2780]: Connection closed by 127.0.0.1 port 5469
mars 15 18:01:51 ubuntu: sshd[2863]: Connection closed by 127.0.0.1 port 4470
mars 15 18:06:51 ubuntu: sshd[2900]: Connection closed by 127.0.0.1 port 5206
mars 15 18:16:51 ubuntu: sshd[3067]: Connection closed by 127.0.0.1 port 3582
mars 15 23:37:33 ubuntu: sshd[4027]: Connection closed by 127.0.0.1 port 6071
mars 15 23:37:37 ubuntu: sshd[4024]: Accepted password for evenort12 from 192.
mars 15 23:37:37 ubuntu: sshd[4024]: pam_unix(sshd:session): session opened for
mars 15 23:47:33 ubuntu: sshd[4281]: Connection closed by 127.0.0.1 port 5218
evenort12@ubuntu:~$ sudo cat /etc/ssh/sshd_config
sudo: unable to resolve host ubuntu: Temporary failure in name resolution

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
```

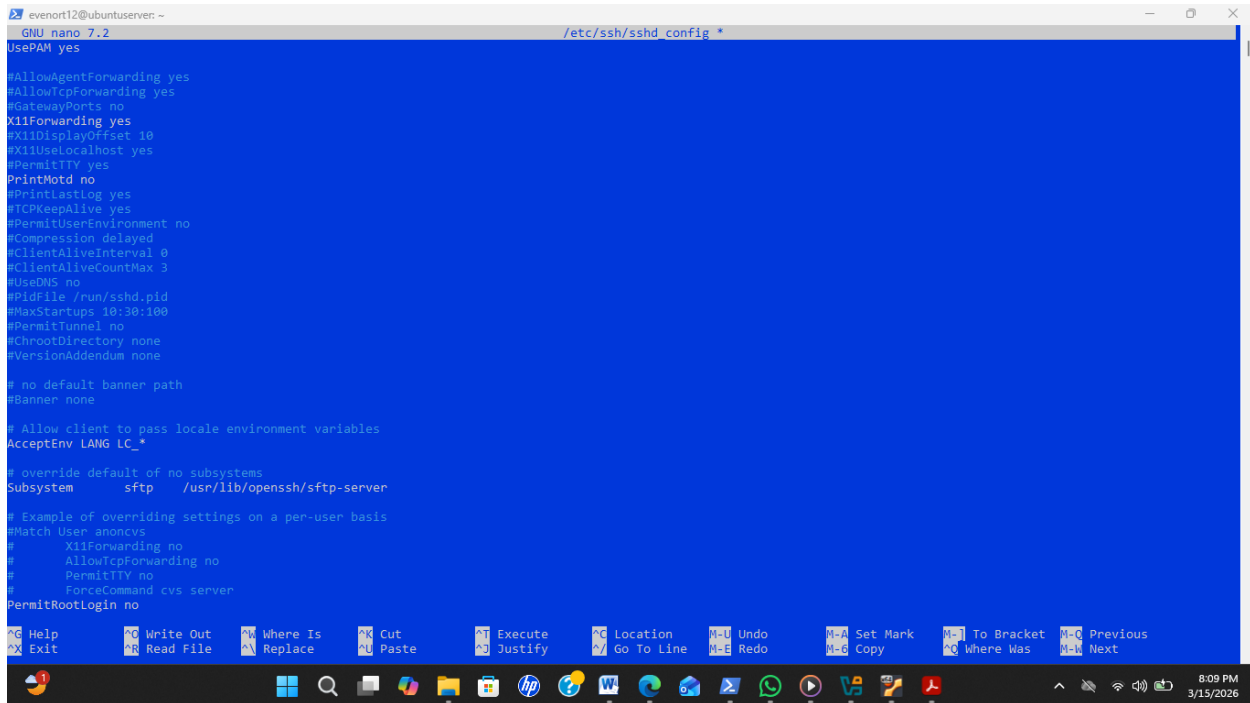
Figure 15 : Cette image confirme le bon fonctionnement du service SSH et présente sa configuration actuelle.

7. Désactiver la connexion root

a) Modifier le fichier : sudo nano /etc/ssh/sshd_config

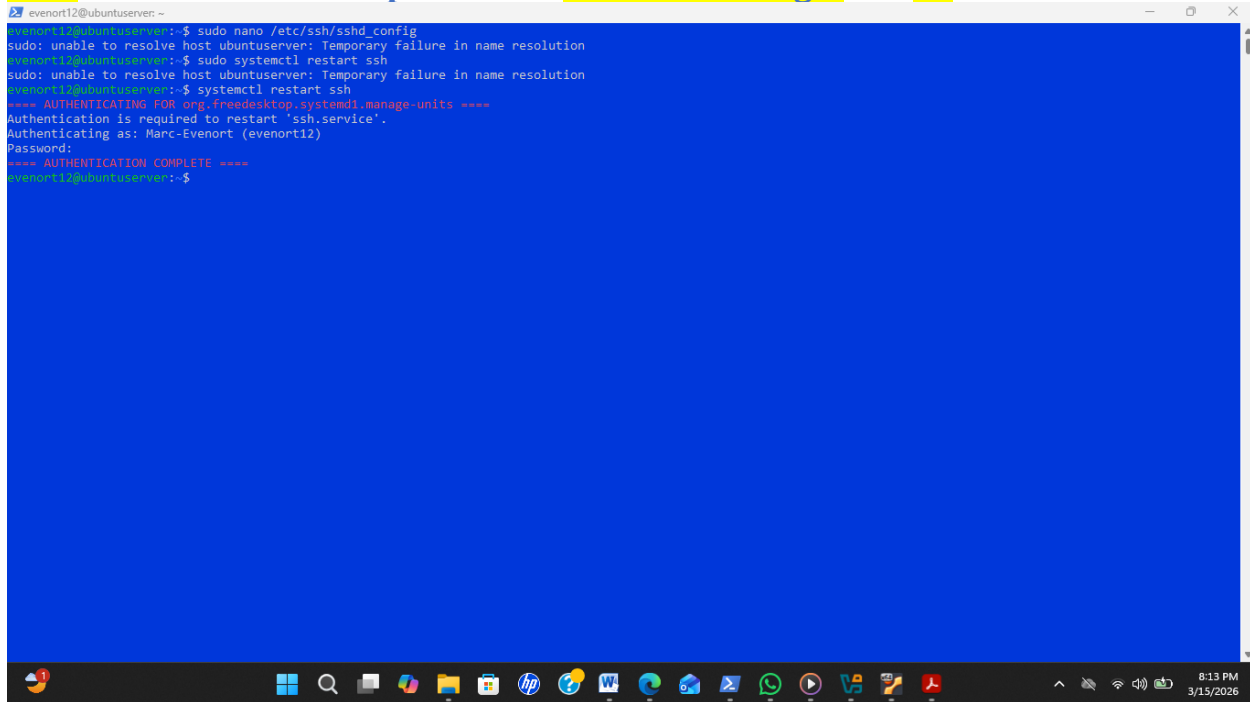
b) Changer : PermitRootLogin no

c) Redémarrer le service : sudo systemctl restart ssh



```
evenort12@ubuntu: ~  
GNU nano 7.2 /etc/ssh/sshd_config  
UsePAM yes  
  
#AllowAgentForwarding yes  
#AllowTcpForwarding yes  
#GatewayPorts no  
X11Forwarding yes  
#X11DisplayOffset 10  
#X11UseLocalhost yes  
#PermitTTY yes  
PrintMotd no  
#PrintLastLog yes  
#TCPKeepAlive yes  
#PermitUserEnvironment no  
#Compression delayed  
#ClientAliveInterval 0  
#ClientAliveCountMax 3  
#UseDNS no  
#PidFile /run/sshd.pid  
#MaxStartups 10:30:100  
#PermitTunnel no  
#ChrootDirectory none  
#VersionAddendum none  
  
# no default banner path  
#Banner none  
  
# Allow client to pass locale environment variables  
AcceptEnv LANG LC_*  
  
# override default of no subsystems  
Subsystem sftp /usr/lib/openssh/sftp-server  
  
# Example of overriding settings on a per-user basis  
#Match User anoncvs  
#    X11Forwarding no  
#    AllowTcpForwarding no  
#    PermitTTY no  
#    ForceCommand cvs server  
PermitRootLogin no  
  
^G Help ^O Write Out ^W Where Is ^K Cut ^X Execute ^C Location ^M-U Undo ^M-A Set Mark ^M-I To Bracket ^M-C Previous  
^X Exit ^R Read File ^N Replace ^U Paste ^J Justify ^V Go To Line ^M-E Redo ^M-G Copy ^M-W Where Was ^M-N Next
```

Figure 16 : Cette image illustre l'édition du fichier de configuration du service SSH afin de modifier le paramètre PermitRootLogin sur no



```
evenort12@ubuntu: ~  
evenort12@ubuntu:~$ sudo nano /etc/ssh/sshd_config  
sudo: unable to resolve host ubuntu: Temporary failure in name resolution  
evenort12@ubuntu:~$ sudo systemctl restart ssh  
sudo: unable to resolve host ubuntu: Temporary failure in name resolution  
evenort12@ubuntu:~$ systemctl restart ssh  
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====  
Authentication is required to restart 'ssh.service'.  
Authenticating as: Marc-Evenort (evenort12)  
Password:  
==== AUTHENTICATION COMPLETE ====  
evenort12@ubuntu:~$
```

Figure 17 : Cette image atteste que la permission root n'a pas été accordée et que le redémarrage du service SSH a été effectué correctement.

8. Authentification par clé SSH

a) Sur la machine cliente :

Créer une clé : ssh-keygen

b) Copier la clé sur le serveur :

ssh-copy-id user@IP_SERVEUR

c) Tester la connexion : ssh evenort12@192.168.56.101

```
evenort12@ubuntu:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/evenort12/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/evenort12/.ssh/id_ed25519
Your public key has been saved in /home/evenort12/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:q0x3NqOfz1FF3yEfm+t00PxGRQ0M7dugfNscCnn7ku0 evenort12@ubuntu:~
The key's randomart image is:
+--[ED25519 256]--+
|  . o  .+o+      |
| o B  .o .      |
| B + ...        |
| o = o .        |
| . S O + o      |
| .. + + o o .   |
| o . o = o o    |
| . .o=* + o .   |
| ...*++E        |
+----[SHA256]-----+
evenort12@ubuntu:~$ ssh-copy-id evenort12@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/evenort12/.ssh/id_ed25519.pub"
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ED25519 key fingerprint is SHA256:hZcPxbTDia6y4zJ51X+EY151wC3UynHkng2/RPnrDz0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
evenort12@192.168.56.101's password:
Permission denied, please try again.
evenort12@192.168.56.101's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'evenort12@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.

evenort12@ubuntu:~$ ssh evenort12@192.168.56.101
Enter passphrase for key '/home/evenort12/.ssh/id_ed25519':
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

* Documentation:  https://help.ubuntu.com
```

Figure 18: Cette image montre la mise en place et la vérification de l'authentification par clé SSH pour sécuriser l'accès au serveur.

```
evenort12@ubuntu:~$ ssh evenort12@192.168.56.101
Now try logging into the machine, with: "ssh 'evenort12@192.168.56.101'"
and check to make sure that only the key(s) you wanted were added.

evenort12@ubuntu:~$ ssh evenort12@192.168.56.101
Enter passphrase for key '/home/evenort12/.ssh/id_ed25519':
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of lun. 16 mars 2026 01:09:07 UTC

System load: 0.02          Memory usage: 8%          Processes: 187
Usage of /: 53.0% of 11.21GB Swap usage: 0%          Users logged in: 1

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

La maintenance de sécurité étendue pour Applications n'est pas activée.

48 mises à jour peuvent être appliquées immédiatement.
33 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

1 mise à jour de sécurité supplémentaire peut être appliquée avec ESM Apps
En savoir plus sur l'activation du service ESM Apps at https://ubuntu.com/esm

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Mon Mar 16 00:54:42 2026 from 192.168.56.101
evenort12@ubuntu:~$ ls -l ~/.ssh
total 24
-rw-r----- 1 evenort12 evenort12 104 mars 16 01:07 authorized_keys
-rw-r----- 1 evenort12 evenort12 464 mars 16 01:05 id_ed25519
-rw-r----- 1 evenort12 evenort12 104 mars 16 01:05 id_ed25519.pub
-rw-r----- 1 evenort12 evenort12 978 mars 16 01:07 known_hosts
-rw-r----- 1 evenort12 evenort12 142 mars 16 01:07 known_hosts.old
drwx----- 2 evenort12 evenort12 4096 mars 6 05:01 ssh-copy-id.506Ke8Hidf
evenort12@ubuntu:~$
```

Figure 19: Cette image montre l'exécution de la commande `ls -l ~/.ssh` pour lister le contenu du répertoire `ssh`.

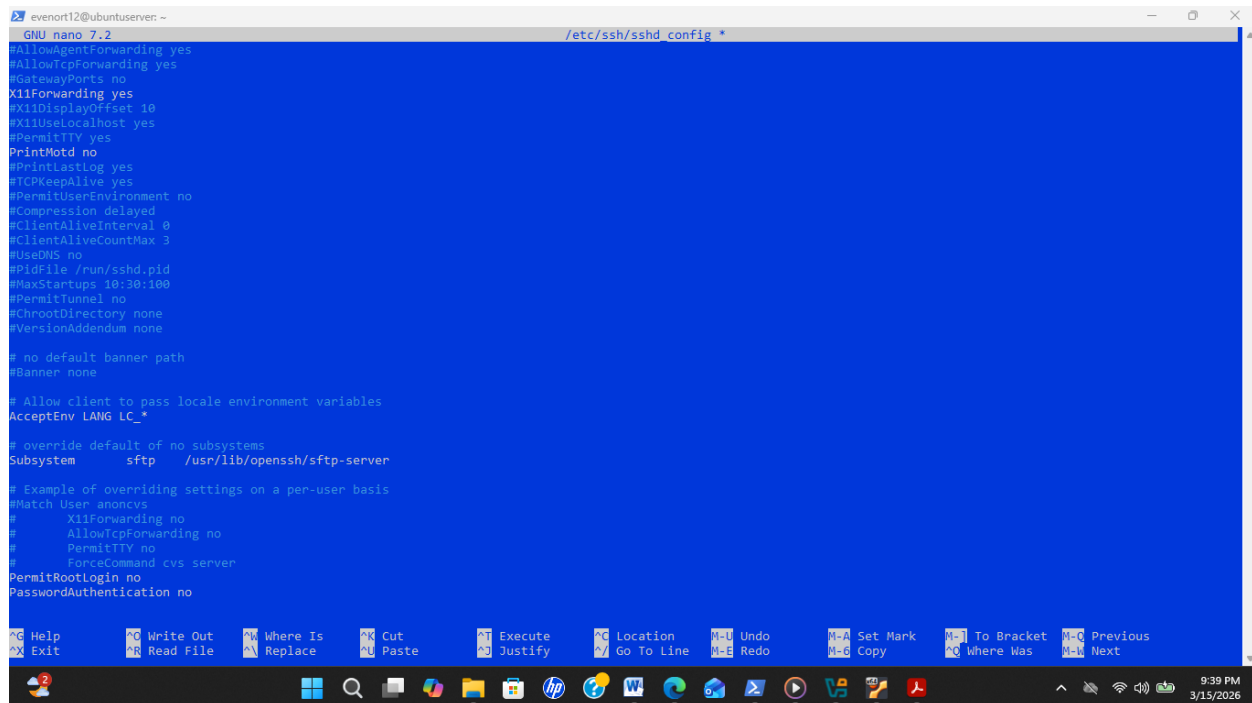
9. Désactiver l'authentification par mot de passe

a) Modifier : `sudo nano /etc/ssh/sshd_config`

 PasswordAuthentication no

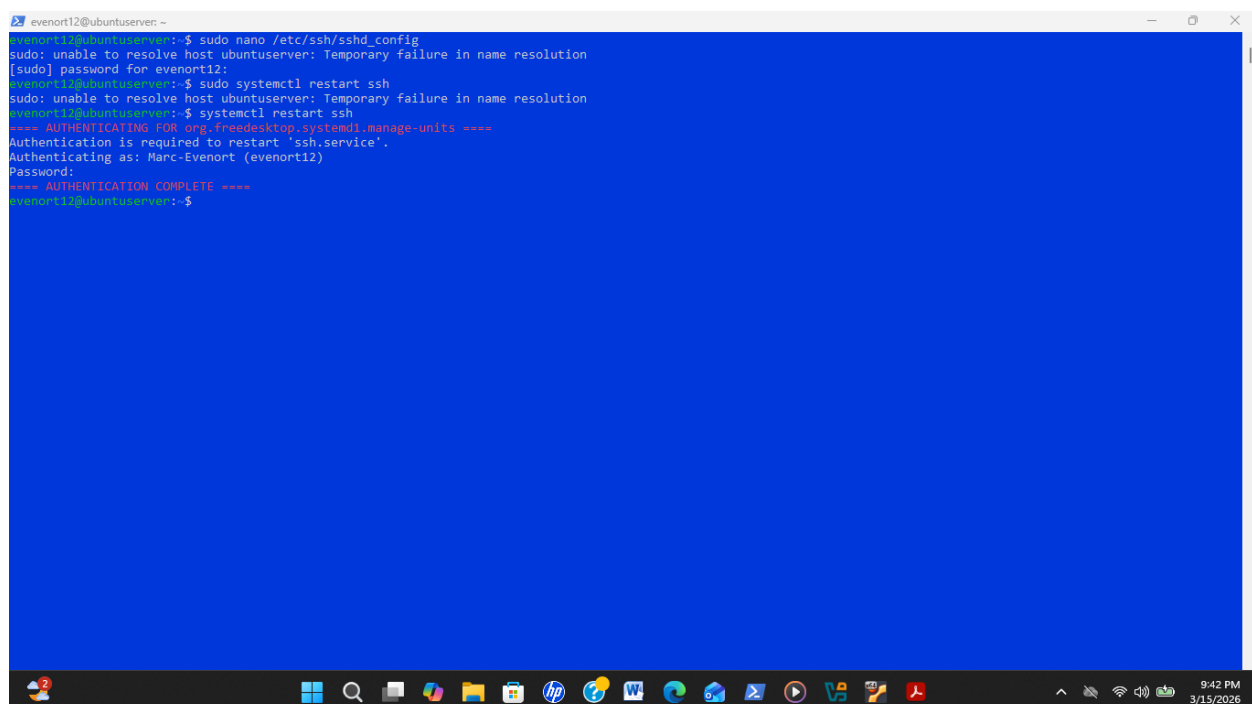
 PermitRootLogin no

b) `sudo systemctl restart ssh`



```
evenort12@ubuntu:server ~  
GNU nano 7.2 /etc/ssh/sshd_config  
#AllowAgentForwarding yes  
#AllowTcpForwarding yes  
#GatewayPorts no  
X11Forwarding yes  
#X11DisplayOffset 10  
#X11UseLocalhost yes  
#PermitTTY yes  
PrintMotd no  
#PrintMotd yes  
#TCPKeepAlive yes  
#PermitUserEnvironment no  
#Compression delayed  
#ClientAliveInterval 0  
#ClientAliveCountMax 3  
#UseDNS no  
#PidFile /run/sshd.pid  
#MaxStartups 10:30:100  
#PermitTunnel no  
#ChrootDirectory none  
#VersionAddendum none  
  
# no default banner path  
#Banner none  
  
# Allow client to pass locale environment variables  
AcceptEnv LANG LC_*  
  
# override default of no subsystems  
Subsystem sftp /usr/lib/openssh/sftp-server  
  
# Example of overriding settings on a per-user basis  
#Match User anoncvs  
#    X11Forwarding no  
#    AllowTcpForwarding no  
#    PermitTTY no  
#    ForceCommand cvs server  
PermitRootLogin no  
PasswordAuthentication no
```

Figure 20: Cette image montre la modification du fichier de configuration SSH pour renforcer la sécurité en désactivant l'authentification par mot de passe **PasswordAuthentication no**.



```
evenort12@ubuntu:server ~  
evenort12@ubuntu:server:~$ sudo nano /etc/ssh/sshd_config  
sudo: unable to resolve host ubuntu:server: Temporary failure in name resolution  
[sudo] password for evenort12:  
evenort12@ubuntu:server:~$ sudo systemctl restart ssh  
sudo: unable to resolve host ubuntu:server: Temporary failure in name resolution  
evenort12@ubuntu:server:~$ systemctl restart ssh  
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====  
Authentication is required to restart 'ssh.service'.  
Authenticating as: Marc-Evenort (evenort12)  
Password:  
==== AUTHENTICATION COMPLETE ====  
evenort12@ubuntu:server:~$
```

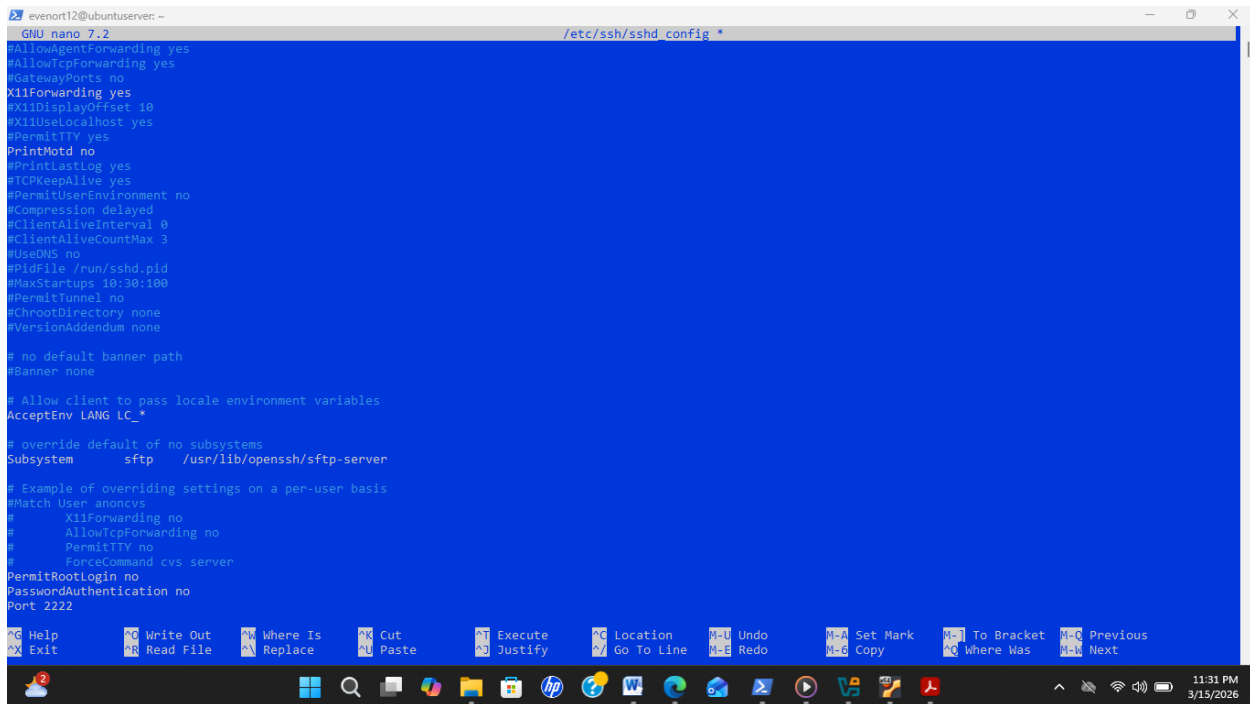
Figure 21: Cette image montre le redémarrage du service SSH (**sudo systemctl restart ssh**) pour appliquer les modifications de configuration.

10. Changer le port SSH

a) **Modifier** : `sudo nano /etc/ssh/sshd_config`

Port 2222

c) **Connexion** : `ssh -p 2222 evenort12@192.168.56.101`



```
evenort12@ubuntu: ~  
GNU nano 7.2 /etc/ssh/sshd_config *  
#AllowAgentForwarding yes  
#AllowTcpForwarding yes  
#GatewayPorts no  
X11Forwarding yes  
#X11DisplayOffset 10  
#X11UseLocalhost yes  
#PermitTTY yes  
PrintMotd no  
#PrintLastLog yes  
#TCPKeepAlive yes  
#PermitUserEnvironment no  
#Compression delayed  
#ClientAliveInterval 0  
#ClientAliveCountMax 3  
#UseDNS no  
#PidFile /run/sshd.pid  
#MaxStartups 10:30:100  
#PermitTunnel no  
#ChrootDirectory none  
#VersionAddendum none  
  
# no default banner path  
#Banner none  
  
# Allow client to pass locale environment variables  
AcceptEnv LANG LC_*  
  
# override default of no subsystems  
Subsystem sftp /usr/lib/openssh/sftp-server  
  
# Example of overriding settings on a per-user basis  
#Match User anoncvs  
#    X11Forwarding no  
#    AllowTcpForwarding no  
#    PermitTTY no  
#    ForceCommand cvs server  
PermitRootLogin no  
PasswordAuthentication no  
Port 2222  
  
^G Help ^O Write Out ^W Where Is ^K Cut ^J Execute ^C Location ^M-U Undo ^M-A Set Mark ^M-I To Bracket ^M-Q Previous  
^X Exit ^R Read File ^N Replace ^P Paste ^_ Go To Line ^M-E Redo ^M-C Copy ^M-] Where Was ^M-N Next
```

Figure 22: Cette image illustre la modification du fichier de configuration SSH pour changer le port par défaut de 22 à 2222, renforçant ainsi la sécurité du serveur.


```
evenort12@ubuntu:~$ nano /etc/ssh/sshd_config
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/ssh.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
#Port 22
Port 2222
PermitRootLogin no
PasswordAuthentication no

evenort12@ubuntu:~$ sudo systemctl restart ssh
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
evenort12@ubuntu:~$ sudo ss -tlnp | grep ssh
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*      users:((("sshd",pid=8609,fd=3),("systemd",pid=1,fd=176))

LISTEN 0      4096      [::]:22        [::]:*        users:((("sshd",pid=8609,fd=4),("systemd",pid=1,fd=177))

evenort12@ubuntu:~$ sudo ufw allow 2222/tcp
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
Skipping adding existing rule
Skipping adding existing rule (v6)
evenort12@ubuntu:~$ sudo ufw status
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
Status: active

To Action From
--
22 ALLOW Anywhere
80 ALLOW Anywhere
443 ALLOW Anywhere
2222/tcp ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)
2222/tcp (v6) ALLOW Anywhere (v6)

evenort12@ubuntu:~$ sudo ss -tlnp | grep ssh
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*      users:((("sshd",pid=8609,fd=3),("systemd",pid=1,fd=176))

LISTEN 0      4096      [::]:22        [::]:*        users:((("sshd",pid=8609,fd=4),("systemd",pid=1,fd=177))

evenort12@ubuntu:~$ sudo sshd -t
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
evenort12@ubuntu:~$ sudo systemctl restart ssh
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
evenort12@ubuntu:~$ sudo ss -tlnp | grep ssh
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
```

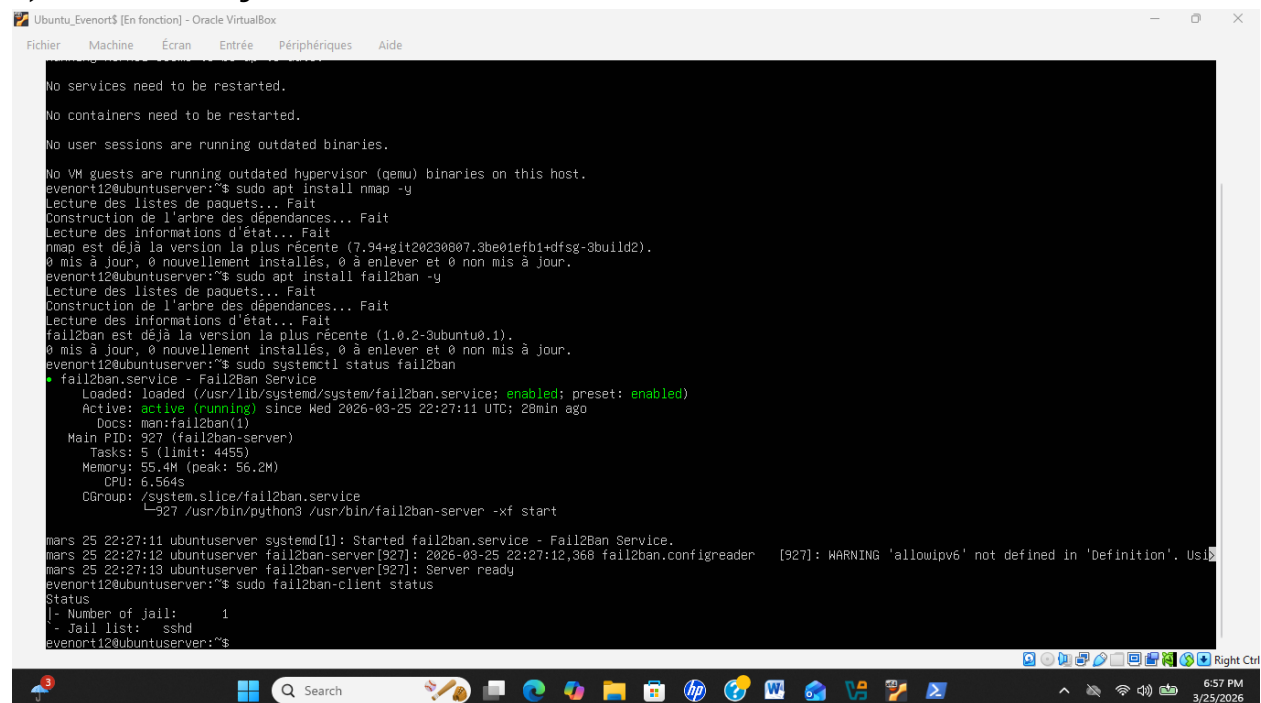
Figure 23, 24: Ces images illustrent la configuration des permissions permettant l'accès au port 2222.

11. Protection contre brute force

a) **Installer** : `sudo apt install fail2ban -y`

b) **Vérifier** : `sudo systemctl status fail2ban`

c) **Afficher les jails actives** : `sudo fail2ban-client status`



```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
evenort12@ubuntu:~$ sudo apt install nmap -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
nmap est déjà la version la plus récente (7.94+git20230807.3be01efb1+dfsg-3build2).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
evenort12@ubuntu:~$ sudo apt install fail2ban -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
fail2ban est déjà la version la plus récente (1.0.2-3ubuntu0.1).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
evenort12@ubuntu:~$ sudo systemctl status fail2ban
* fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-03-25 22:27:11 UTC; 20min ago
     Docs: man:fail2ban(1)
    Main PID: 927 (fail2ban-server)
      Tasks: 5 (limit: 4455)
     Memory: 55.4M (peak: 56.2M)
        CPU: 6.564s
    CGroup: /system.slice/fail2ban.service
            └─927 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

mars 25 22:27:11 ubuntu:systemd[1]: Started fail2ban.service - Fail2Ban Service.
mars 25 22:27:12 ubuntu:fail2ban-server[927]: 2026-03-25 22:27:12,368 fail2ban.configreader [927]: WARNING 'allowipv6' not defined in 'Definition'. Usi
mars 25 22:27:13 ubuntu:fail2ban-server[927]: Server ready
evenort12@ubuntu:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
  - Jail list:         sshd
evenort12@ubuntu:~$
```

Figure 25: Sur cette image, après avoir installé Fail2ban, on constate que le service Fail2ban fonctionne correctement ainsi que la commande fail2ban-client.

12. Tests de sécurité

a) **Vérifier les ports ouverts** :

`ss -tuln`

`nmap 192.168.56.101`

```
Ubuntu_Evenort$ [En fonction] - Oracle VirtualBox
Fichier  Machine  Écran  Entrée  Périphériques  Aide

ED25519 key fingerprint is SHA256:+2+LN07CDA9Q4a9IMFegqtoCxAptgOQJGbmR/ttu04.
This host key is known by the following other names/addresses:
  /usr/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.101' (ED25519) to the list of known hosts.
Enter passphrase for key '/home/evenort12/.ssh/id_ed25519':
Last login: Wed Mar 25 04:30:11 2026 from 192.168.56.1
evenort12@ubuntu:~$ sudo apt install fail2ban
[sudo] password for evenort12:
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
fail2ban est déjà la version la plus récente (1.0.2-3ubuntu0.1).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
evenort12@ubuntu:~$ sudo systemctl status fail2ban
fail2ban.service - Fail2Ban Service
Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
Active: active (running) since Wed 2026-03-25 04:22:30 UTC; 24min ago
Docs: man:fail2ban(1)
Main PID: 939 (fail2ban-server)
Tasks: 5 (limit: 4455)
Memory: 52.1M (peak: 52.9M)
CPU: 7.033s
CGroup: /system.slice/fail2ban.service
└─939 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

mars 25 04:22:30 ubuntu:systemd[1]: Started fail2ban.service - Fail2Ban Service.
mars 25 04:22:30 ubuntu:fail2ban-server[939]: 2026-03-25 04:22:30,776 fail2ban.configreader [939]: WARNING 'allowipv6' not defined in 'Definition'. Usi
evenort12@ubuntu:~$ ss -tlnl
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
udp        UNCONN    0           0           127.0.0.54:53            0.0.0.0:*
udp        UNCONN    0           0           127.0.0.53:10:53        0.0.0.0:*
udp        UNCONN    0           0           192.168.56.101:53:68    0.0.0.0:*
udp        UNCONN    0           0           0.0.0.0:111             0.0.0.0:*
udp        UNCONN    0           0           127.0.0.1:161           0.0.0.0:*
udp        UNCONN    0           0           [::]:111                [::]:*
udp        UNCONN    0           0           [::1]:161               [::]:*
tcp        LISTEN    0           4096        0.0.0.0:111             0.0.0.0:*
tcp        LISTEN    0           511         0.0.0.0:80              0.0.0.0:*
```

Figure 26: Cette image présente les résultats des tests de sécurité réalisés avec la commande **ss -tlnl**, permettant de vérifier les ports ouverts et les services actifs.

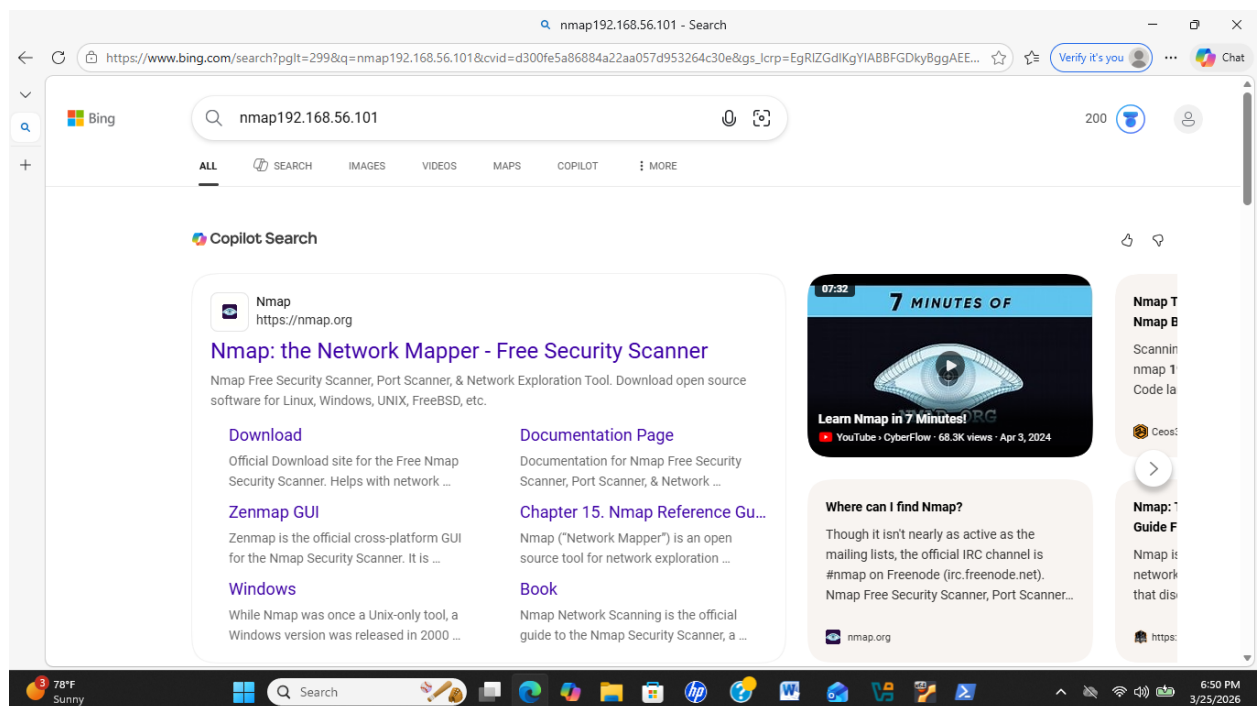


Figure 27: Cette image montre les résultats de l'identification avec **Nmap**

1. Différence entre firewall stateful et stateless

- **Firewall stateless :**
Analyse chaque paquet individuellement sans tenir compte des connexions précédentes.
Il applique seulement des règles simples (adresse IP, port, protocole).
- **Firewall stateful :**
Analyse les paquets en tenant compte de l'état de la connexion (nouvelle connexion, connexion établie, etc.).
Il est plus intelligent et plus sécurisé.

2. Avantages de nftables par rapport à iptables

- Configuration plus simple et plus moderne
- Performance plus élevée
- Permet de gérer **IPv4 et IPv6** dans une seule configuration
- Syntaxe plus claire et plus flexible
- Remplace progressivement iptables dans les systèmes Linux récents.

3. Pourquoi toujours autoriser SSH avant d'activer un firewall ?

Parce que si le firewall bloque le **port SSH (22)**, l'administrateur **perdra l'accès au serveur à distance**.

Résultat :

Il faudra accéder physiquement au serveur pour corriger la configuration.

Donc on ajoute toujours une règle comme :

allow SSH

avant d'activer le firewall.

4. Pourquoi désactiver la connexion root ?

R) Pour des raisons de **sécurité**.

- Le compte root est **la cible principale des hackers**
- Si quelqu'un obtient l'accès root → **contrôle total du système**

Bonne pratique :

- Se connecter avec un utilisateur normal

- Utiliser **sudo** pour les privilèges administrateur.

5. Avantages de changer le port SSH

Changer le port SSH (ex : 22 → 2222) permet :

- Réduire les **attaques automatiques**
- Diminuer les **tentatives de brute force**
- Améliorer légèrement la sécurité

Ce n'est pas une protection totale, mais une mesure supplémentaire.

6. Quel est le rôle de Fail2Ban ?

Fail2Ban est un outil de sécurité qui :

- Surveille les **logs du système**
- Détecte les **tentatives de connexion échouées**
- **Bloque automatiquement l'adresse IP** de l'attaquant

Remarque: Si quelqu'un échoue **5 fois à se connecter en SSH**, Fail2Ban **bloque son IP pendant un certain temps.**

Conclusion

Au terme de ce TP, nous avons pu installer et configurer les pare-feu, vérifier les règles en place, tester les ports ouverts et assurer la protection du service SSH grâce à Fail2Ban. Ces opérations confirment que le serveur est correctement sécurisé et que les services essentiels fonctionnent de manière optimale, renforçant ainsi la sécurité globale du système.